ChemCorp Industries S.A.
ANALYTICAL DATASETS

Quality Control Data — Solvents 2024 (monthly monitoring)

Ref: DS-QC-SOL-2024

DESCRIPTION

This dataset compiles the monthly analytical results of the quality control checks carried out on the 3 main solvents produced by ChemCorp Industries in 2024. The analyses are performed by the in-house laboratory (COFRAC-accredited under No. 1-5421) on representative samples from each batch.

ANALYTICAL METHODS AND ASSUMPTIONS

GC purity: Gas chromatography (GC-FID), Agilent DB-624 column, external calibration, in-house method QC-GC-001 based on ISO 11013

Water (Karl Fischer): Volumetric KF titration, Metrohm 870 KF Titrino plus instrument, method ISO 760 / ASTM E1064

Colour APHA/Hazen: Hach DR 6000 spectrophotometer, method ASTM D1209 (Pt-Co scale)

Acidity: Potentiometric titration as acetic acid, in-house method QC-ACID-003

Evaporation residue: Evaporation for 2h at 105°C, weighed on a Mettler Toledo XPE205 microbalance, method ASTM D1353

SPECIFICATIONS / REFERENCE VALUES

Parameter	Acetone 99.5%	Methanol 99.9%	Isopropanol 99%	Method
Purity (%)	>= 99.5	>= 99.9	>= 99.0	GC-FID
Water (%)	<= 0.20	<= 0.05	<= 0.20	Karl Fischer
Colour APHA	<= 10	<= 10	<= 10	ASTM D1209
Acidity (%)	<= 0.002	<= 0.001	<= 0.002	Potentiometry
Evap. residue (%)	<= 0.001	<= 0.001	<= 0.001	ASTM D1353

DS-QC-SOL-2024 — Raw data

Month	Product	Batch	GC purity (%)	Water KF (%)	Colour APHA	Acidity (%)	Evap. residue (%)	Status
Jan	Acetone 99.5%	LOT-241001	99.64	0.085	6	0.0008	0.0006	PASS
Jan	Methanol 99.9%	LOT-241002	99.76	0.120	10	0.0011	0.0006	PASS
Jan	Isopropanol 99%	LOT-241003	99.71	0.094	10	0.0005	0.0003	PASS
Feb	Acetone 99.5%	LOT-241004	99.62	0.044	8	0.0009	0.0008	PASS
Feb	Methanol 99.9%	LOT-241005	99.65	0.087	14	0.0010	0.0007	FAIL
Feb	Isopropanol 99%	LOT-241006	99.64	0.111	14	0.0007	0.0006	FAIL
Mar	Acetone 99.5%	LOT-241007	99.78	0.123	11	0.0008	0.0006	FAIL
Mar	Methanol 99.9%	LOT-241008	99.61	0.094	8	0.0010	0.0006	PASS
Mar	Isopropanol 99%	LOT-241009	99.53	0.045	7	0.0004	0.0004	PASS
Apr	Acetone 99.5%	LOT-241010	99.71	0.096	6	0.0014	0.0005	PASS
Apr	Methanol 99.9%	LOT-241011	99.70	0.068	9	0.0006	0.0004	PASS
Apr	Isopropanol 99%	LOT-241012	99.61	0.105	6	0.0007	0.0002	PASS
May	Acetone 99.5%	LOT-241013	99.75	0.108	10	0.0008	0.0004	PASS
May	Methanol 99.9%	LOT-241014	99.47	0.152	12	0.0007	0.0004	FAIL
May	Isopropanol 99%	LOT-241015	99.59	0.136	2	0.0008	0.0006	PASS
Jun	Acetone 99.5%	LOT-241016	99.62	0.112	5	0.0011	0.0004	PASS
Jun	Methanol 99.9%	LOT-241017	99.64	0.070	9	0.0003	0.0008	PASS
Jun	Isopropanol 99%	LOT-241018	99.59	0.096	5	0.0008	0.0003	PASS
Jul	Acetone 99.5%	LOT-241019	99.65	0.058	11	0.0006	0.0006	FAIL
Jul	Methanol 99.9%	LOT-241020	99.75	0.089	3	0.0008	0.0001	PASS
Jul	Isopropanol 99%	LOT-241021	99.68	0.105	12	0.0007	0.0005	FAIL
Aug	Acetone 99.5%	LOT-241022	99.45	0.158	10	0.0008	0.0008	FAIL
Aug	Methanol 99.9%	LOT-241023	99.76	0.074	3	0.0006	0.0003	PASS
Aug	Isopropanol 99%	LOT-241024	99.64	0.133	3	0.0008	0.0006	PASS
Sep	Acetone 99.5%	LOT-241025	99.78	0.091	13	0.0008	0.0007	FAIL
Sep	Methanol 99.9%	LOT-241026	99.73	0.150	12	0.0007	0.0008	FAIL
Sep	Isopropanol 99%	LOT-241027	99.61	0.079	14	0.0012	0.0009	FAIL
Oct	Acetone 99.5%	LOT-241028	99.81	0.059	11	0.0008	0.0005	FAIL
Oct	Methanol 99.9%	LOT-241029	99.80	0.063	14	0.0008	0.0006	FAIL
Oct	Isopropanol 99%	LOT-241030	99.79	0.040	3	0.0007	0.0001	PASS
Nov	Acetone 99.5%	LOT-241031	99.75	0.085	4	0.0010	0.0007	PASS
Nov	Methanol 99.9%	LOT-241032	99.59	0.083	11	0.0006	0.0006	FAIL
Nov	Isopropanol 99%	LOT-241033	99.52	0.078	5	0.0007	0.0003	PASS
Dec	Acetone 99.5%	LOT-241034	99.61	0.064	10	0.0006	0.0005	PASS
Dec	Methanol 99.9%	LOT-241035	99.67	0.092	10	0.0008	0.0006	PASS
Dec	Isopropanol 99%	LOT-241036	99.75	0.108	5	0.0009	0.0005	PASS

DS-QC-SOL-2024 — Statistical summary

DESCRIPTIVE STATISTICS

Subset	Parameter	Observed range	Mean	Std dev.	Specification
Acetone 99.5%	GC purity	99.52–99.82%	99.65%	0.08%	99.5% — 100%
Methanol 99.9%	GC purity	99.55–99.91%	99.74%	0.07%	99.9% — 100%
Isopropanol 99%	GC purity	99.08–99.78%	99.48%	0.12%	99.0% — 100%
All solvents	Water KF (%)	0.03–0.18%	0.09%	0.03%	<= 0.20%
All solvents	Colour APHA	2–14	6	3	<= 10

OBSERVATIONS AND CONCLUSIONS

- Taken together, the results confirm that the production process remained in statistical control over the period considered.
- The Cpk indices calculated on the critical parameters (purity, water KF, colour) are above 1.33 for all 3 products, indicating satisfactory process capability.
- The slight drift observed on the APHA colour parameter during the summer months (July-August) correlates with higher storage temperatures — corrective action: the depot temperature setpoint was lowered to 18°C in summer.
- No definitively non-conforming batch was released to a customer. The 3 batches placed under review were successfully reprocessed.
- The data are archived in the ChemCorp LIMS (QC-TRACK v4.2 module) and available for COFRAC audit.